

École Pour l'Informatique et les Techniques Avancées – EPITA

BSc L1

Course: Introduction to Computer Networks



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Introduction to Computer Networks

Course schedule (tentative)

Check Zeus platform for the date & time	No.	Topics	Check Zeus platform for the duration (hrs)
-	1	Primer, Network protocols, types, topology, architecture	-
-	2	Network models, TCP/IP model, Packet switching	-
-	3	Physical Layer (Function, Signals, Modulation, Multiplexing, Transmission media & Hardware, Optical networks)	-
-	4	Data Link Layer (Function, Framing, Protocols, Flow control, Access control, Error correction, Hardware)	-
-	5	Network Layer (Function, IP addressing and subnets)	-
-	6	Network Layer (Routing algorithms and protocols), Internet Control Message Protocol	-
-	7	Network Layer (IGP & EGP), Autonomous System, Border Gateway Protocol	-
-	8	Transport Layer (Function, Flow and congestion controls, Protocols)	-
-	9	Cross-layer process: Access Control Lists	-

Introduction to Computer Networks

Check Zeus platform for the date & time	No.	Topics	Check Zeus platform for the duration (hrs)
-	10	Application Layer (DNS, SMTP)	-
-	11	Cross-layer process: Network Address Translation	-
	12	Review / Open-session	-
Total			35
-		EXAM	1

GRADING criteria:

Class attendance and responsiveness: 10%

Class assignments (practical work): 40%

Final evaluation (Quiz & Exam): 50%

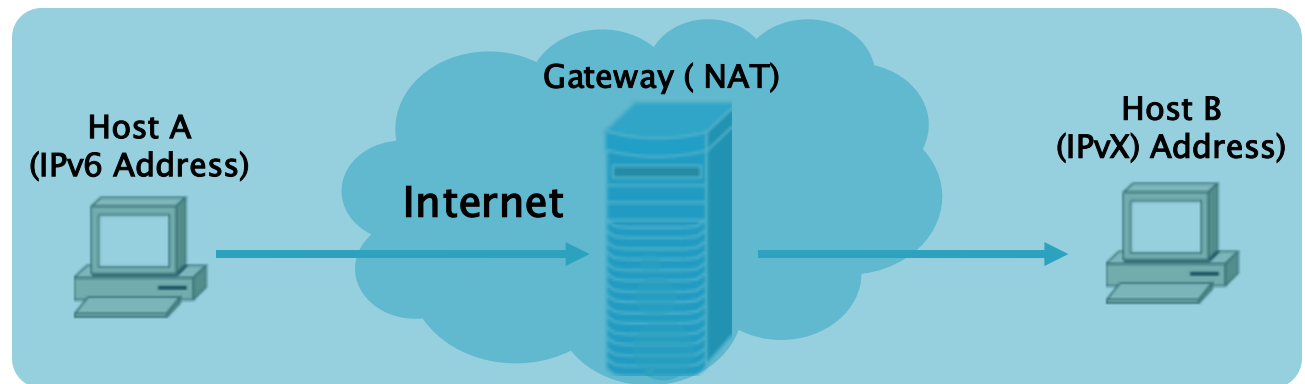
i Check policies in the course outline

Lecture 1 1 Outline

- ▶ **Network Address Translation (NAT)**
 - **Types**
 - **One-to-Many NAT**
 - **Class exercise 17**
- ▶ **Run-down**
 - **Open session**

NAT (Network Address Translation)

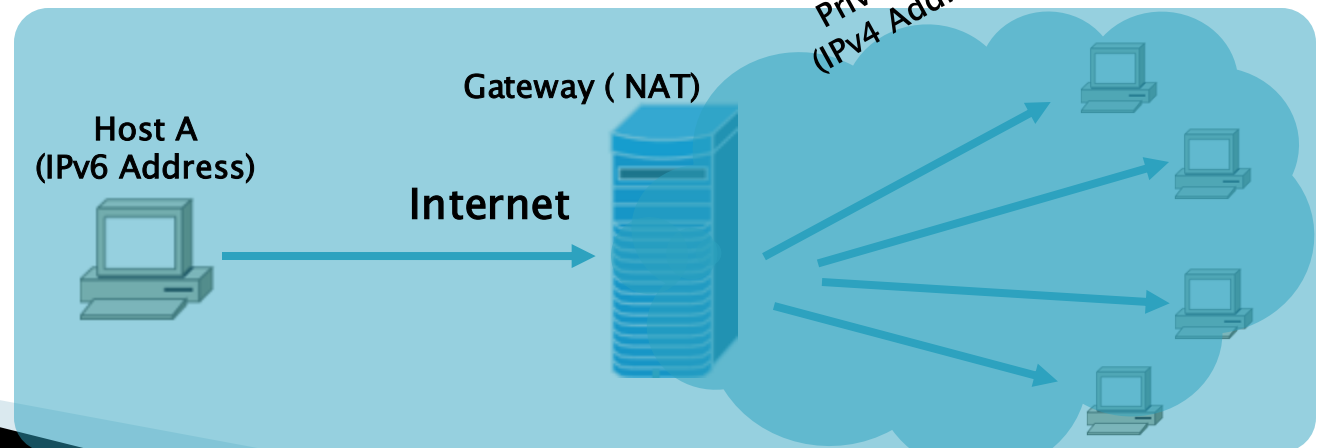
- ▶ A built-in component of most Routers
- ▶ NAT main Types:
 - **One-to-One NAT (a.k.a Basic or Static NAT):**
 - Enables interconnection of two incompatibly addressed (e.g., IPv4 to IPv6 & vice versa) or different assigned IP networks
 - Follows RFC 2663



NAT (cont.)

► NAT main Types:

- **One-to-Many NAT (A.k.a. IP Masquerading / Overload):**
 - IP address space (with many private IP addresses) hidden behind a single public IP address
 - Uses Port Address Translation (PAT): Inside Local and Inside Global PAT table (each local device is mapped with the public IP address using a unique port number)



One-to-Many NAT

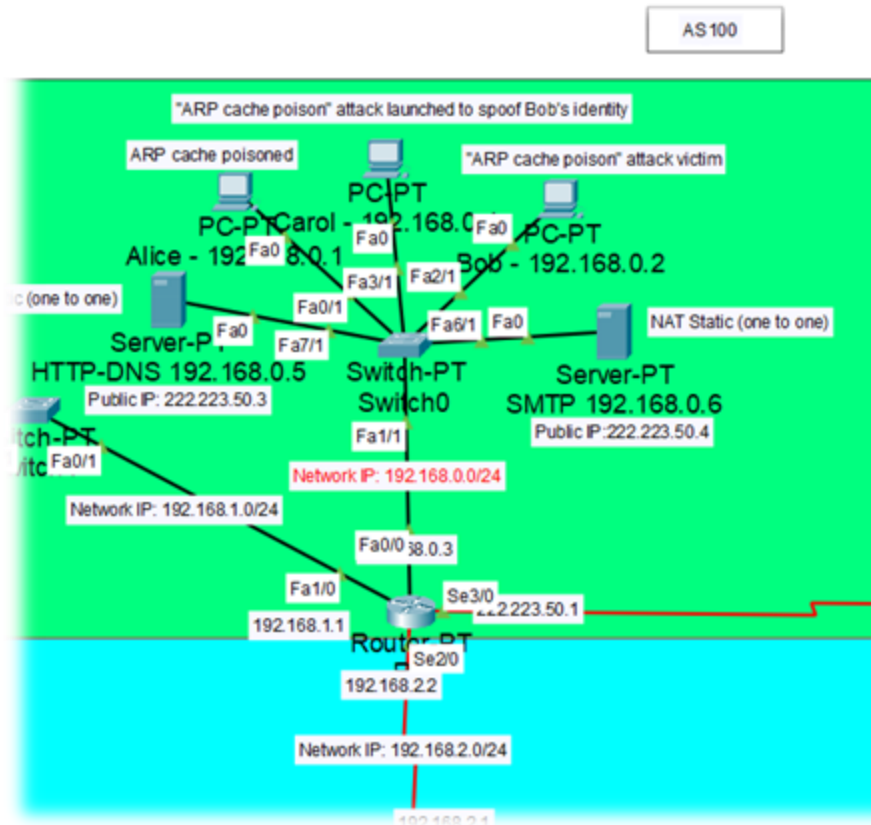
▶ Pro's:

- Provides enhanced security
- Facilitate the process of saving public IPv4 addresses
 - Partial solution for the IPv4 address space exhaustion problem
 - ...

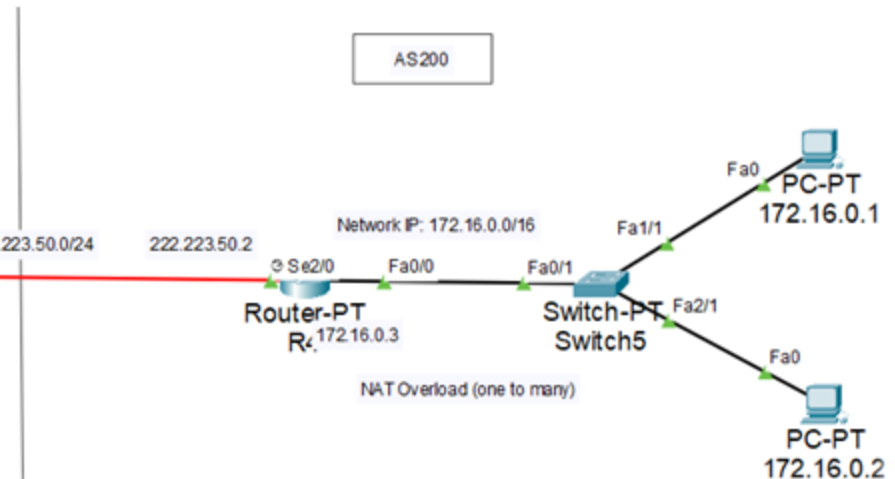
▶ Con's

- Creates overhead/latency
- Difficult to manage with protocols that detect changes in headers (e.g., IPSec, ...)
- ...

Exercise 17: Practical work (1 / 2)



Inside Local: Fa0/0
Inside Global: Se3/2
Outside Local: Fa0/0
Outside Global: Se2/0



Inside Local: Fa0/0
Inside Global: Se2/2
Outside Local: Fa0/0
Outside Global: Se3/0

Exercise 17: Practical work (2/2)

► Using your last cisco packet tracer file:

1. Apply NAT overload in AS200 for NID: 172.16.0.0

```
R# (config-if)#ip nat inside //interface: inside local e.g., Fa0/0
```

```
R# (config-if)#ip nat outside //interface: inside global e.g., Se6/0
```

```
R# (config)# access-list NUM permit IP_OR_Net-ID Wild_Card
```

//Using access list to designate inside local IP address (or 'Interesting traffic ') to be used for dynamic translation for NAT

```
R# (config)# ip NAT inside source list NUM interface INT_Num overload
```

//NATing using the access list for 'local inside' devices with the IP address of 'outside local' interface IP

2. Apply NAT Static on HTTP and Mail server on AS100, and restrict these services for AS 200 only

```
R# (config-if)#ip nat inside //interface: inside local e.g., Fa0/0
```

```
R# (config-if)#ip nat outside //interface: inside global e.g., Se6/0
```

```
R# (config)# ip NAT inside source static Private_IP_Address Public_IP_Address
```

```
R# Show ip nat translations
```

i Static NAT: We can initiate translation from both inside to outside and vice versa. That is not desired in Dynamic NAT and Overload (PAT)

Verify from outside networks using PING or TRACERT/TRACEROUTE.

Save cisco packet tracer file with your 'First_Last-name_Ex-6-17'

Deadline: See 'Teams' Assignment section

Lecture 1 1 Outline

- ▶ Network Address Translation (NAT)
 - Types
 - One-to-Many NAT
 - Class exercise 17
- ▶ Run-down
 - Open session

Lecture 1 1 ends here

- ▶ Course Slides: Go to MS Teams:
'Introduction to Computer Networks – Spring 2026 | BSc'
→ Files section
- ▶ Send your questions by email:
mohammad-salman.nadeem@epita.fr
OR via direct message using MS Teams
- ▶ Thank You!

Open session

- ▶ Practice discussed topics on Network Simulation Program
- ▶ Questions?

General reference list

- ▶ Internet Engineering Task Force. www.ietf.org.
- ▶ Wikipedia contributors. Wikipedia. Retrieved from <https://www.wikipedia.org/>
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- ▶ Cisco Systems. (n.d.). Resolve IP Fragmentation, MTU, MSS, and PMTUD Issues with GRE and IPSEC. Retrieved from <http://www.cisco.com/c/en/us/support/docs/ip/generic-routing-encapsulation/gre/25885-pmtud-ipfrag.html> [Accessed June 1, 2015]
- ▶ Microsoft. (n.d.). IPv6 Technical Reference. Retrieved from [https://technet.microsoft.com/en-us/library/cc738109\(v=ws.10\).aspx](https://technet.microsoft.com/en-us/library/cc738109(v=ws.10).aspx) [Accessed June 1, 2015]
- ▶ Shinder, D. (2011). Networking For Dummies (8th ed.). John Wiley & Sons.
- ▶ Stallings, W. (2013). Data and Computer Communications (10th ed.). Pearson.
- ▶ Peterson, L. L., & Davie, B. S. (2011). Computer Networks: A Systems Approach (5th ed.). Morgan Kaufmann.
- ▶ Postel, J. (1981). Internet Protocol DARPA Internet Program Protocol Specification (RFC 791). Internet Engineering Task Force.